Ch 3.3: Even More Linear Regression

Lecture 8 - CMSE 381

Rachel Roca and Prof. Elizabeth Munch

Michigan State University
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Dept of Computational Mathematics, Science & Engineering

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Announcements

Last time:

- 3.2 Multiple Linear Regression

Announcements:

- Dr. Munch back next week!

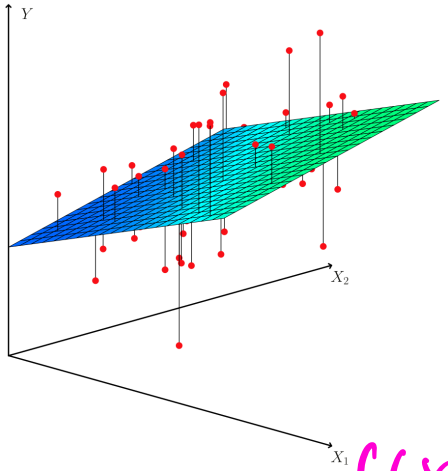
Covered in this lecture

- Extending the linear model with interaction terms
- Hierarchy principle
- Polynomial regression

Section 1

Review from last time

Linear Regression with Multiple Variables : $x_1, \dots, x_p$



$f(x)$

- Predict Y on a multiple variables X

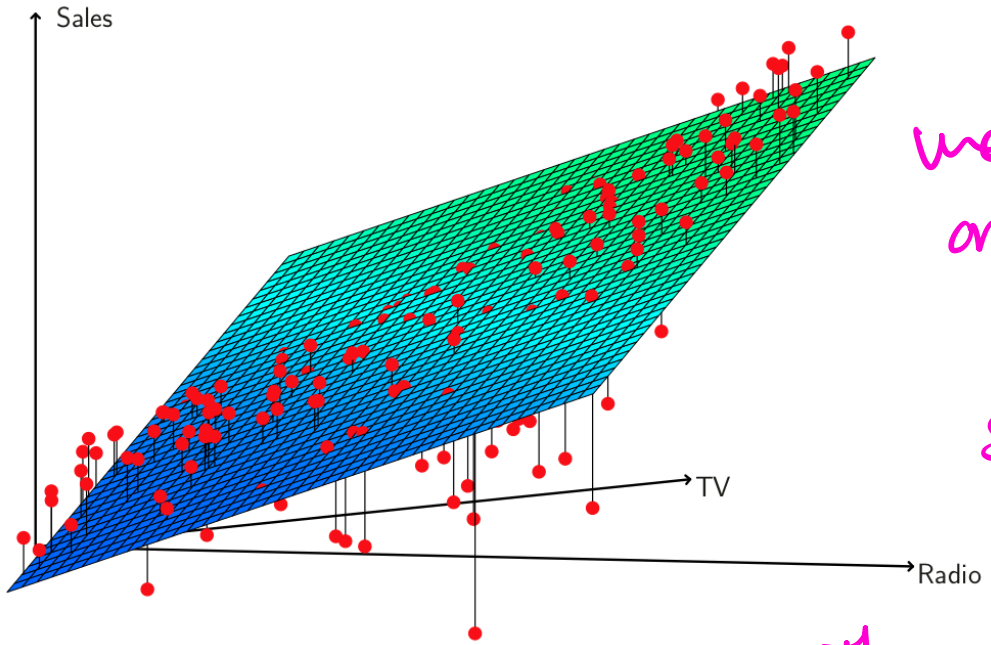
$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots \beta_p x_p + \varepsilon$$

- Find good guesses for $\hat{\beta}_0$, $\hat{\beta}_1, \cdots$.
- $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + \cdots + \hat{\beta}_p x_p$

$\hat{f}(x)$

- $e_i = y_i - \hat{y}_i$ is the ith residual
- RSS $= \sum_i e_i^2$
- RSS is minimized at *least squares coefficient estimates*

If all else fails, look at the data



- Model
  overestimates sales when we spend on TV or Radio only
  underestimates when we split our budget between the two.

There is some interaction between advertising for Radio and TV

Section 2

Extending the linear model

Assumptions so far

Back to our Advertising data set

$$\hat{Y}_{sales} = \beta_0 + \beta_1 \cdot X_{TV} + \beta_2 \cdot X_{radio} + \beta_3 \cdot X_{newspaper}$$

Assumed (implicitly) that the effect on sales by increasing one medium is independent of the others.

What if spending money on radio advertising increases the effectiveness of TV advertising? How do we model it?

Interaction Term

interaction term

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon$$

$$\begin{aligned} Y_{sales} &= \beta_0 + \beta_1 X_{TV} + \beta_2 X_{radio} + (\beta_3 X_{radio}) X_{TV} + \varepsilon \\ &= \beta_0 + (\beta_1 + \beta_3 X_{radio}) X_{TV} + \beta_2 X_{radio} + \varepsilon \\ &= \beta_0 + \tilde{\beta}_1 X_{TV} + \beta_2 X_{radio} + \varepsilon \end{aligned}$$

$\hookrightarrow \tilde{\beta}_1 = \beta + \beta_3 X_{radio}$

$\tilde{\beta}_1$ function of X_{radio}, the association is no longer constant.

Interaction term

β_3: the increase of effectiveness in TV ad. associated w/ one-unit increase in Radio ad.

R^2 w/ interaction term is .968
w/out it is .897

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon$$

	Coefficient	Std. error	t-statistic	p-value
Intercept	6.7502	0.248	27.23	< 0.0001
TV	0.0191	0.002	12.70	< 0.0001
radio	0.0289	0.009	3.24	0.0014
TV×radio	0.0011	0.000	20.73	< 0.0001

$$\begin{aligned} Y_{sales} &= \beta_0 + \beta_1 X_{TV} + \beta_2 X_{radio} + \beta_3 X_{radio} X_{TV} + \varepsilon \\ &= \beta_0 + (\beta_1 + \beta_3 X_{radio}) X_{TV} + \beta_2 X_{radio} + \varepsilon \end{aligned}$$

p-val is low, strong evidence to reject H_0 $\beta_3 \neq 0$.

Interpretation

$\tilde{\beta}_1 = \beta_1 + \beta_3 X_{radio}$

		Coefficient	Std. error	t-statistic	p-value
β_0	Intercept	6.7502	0.248	27.23	< 0.0001
β_1	TV	0.0191	0.002	12.70	< 0.0001
β_2	radio	0.0289	0.009	3.24	0.0014
β_3	TV×radio	0.0011	0.000	20.73	< 0.0001

β_3

- Increasing \$1000 of TV is associated to increased of

$(\beta_1 + \beta_3 X_{radio}) \cdot 1000 = 19 + 1.1 X_{radio}$

$\beta_1 \to 0.0191$, $\beta_3 \to .0011$

- Increasing \$1000 of radio advertising will be associated to

$(\beta_2 + \beta_3 X_{TV}) \cdot 1000 = 29 + 1.1 X_{TV}$

Do the section on “Interaction Terms”

Hierarchy principle

$$\hat{y}_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2$$

main effects

interaction term

Sometimes p-value for interaction term is very small, but associated main effects are not.

The hierarchy principle:

If we include the interaction term in our model, we should include the main effects no matter what

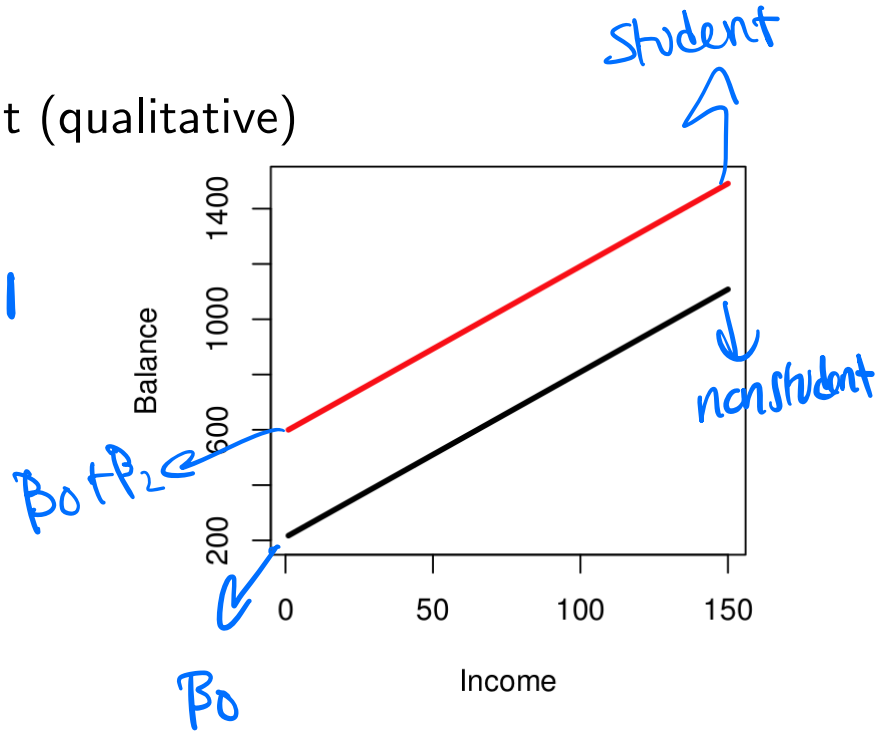
Interaction term for qualitative variables

Without interaction term

For credit data set:
Predict balance using income (quantitative) and student (qualitative)

$$\texttt{balance}_i \approx \beta_0 + \beta_1 \cdot \texttt{income}_i + \begin{cases} \beta_2 & \text{if student} \quad 1 \\ 0 & \text{if not} \quad 0 \end{cases}$$

$$\approx \beta_1 \cdot \texttt{income}_i + \begin{cases} \beta_0 + \beta_2 & \text{if student} \\ \beta_0 & \text{if not} \end{cases}$$



2 parallel lines for our 2 subsets of data. This means the average effect of 1 unit increase in income does not depend on whether the person is student

whether the person is a student.

Interaction term for qualitative variables

With interaction term

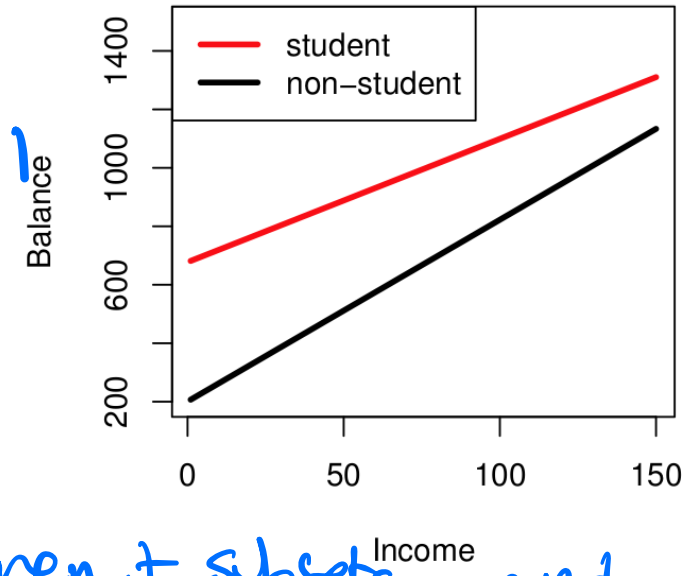
$\text{balance}_i \approx \beta_0 + \beta_1 \text{income} + \beta_2 \text{student} + \beta_3 \text{income} \times \text{student}$

For credit data set:

Predict balance using income (quantitative) and student (qualitative)

$$\texttt{balance}_i \approx \beta_0 + \beta_1 \cdot \texttt{income}_i + \begin{cases} \beta_2 + \beta_3 \cdot \texttt{income}_i & \text{if student } 1 \\ 0 & \text{if not } 0 \end{cases}$$

$$\approx \begin{cases} (\beta_0 + \beta_2) + (\beta_1 + \beta_3) \cdot \texttt{income}_i & \text{if student} \\ \beta_0 + \beta_1 \cdot \texttt{income}_i & \text{if not} \end{cases}$$



fitting 2 different lines to the different subsets, and increases in income are associated with smaller increases in credit card balance among

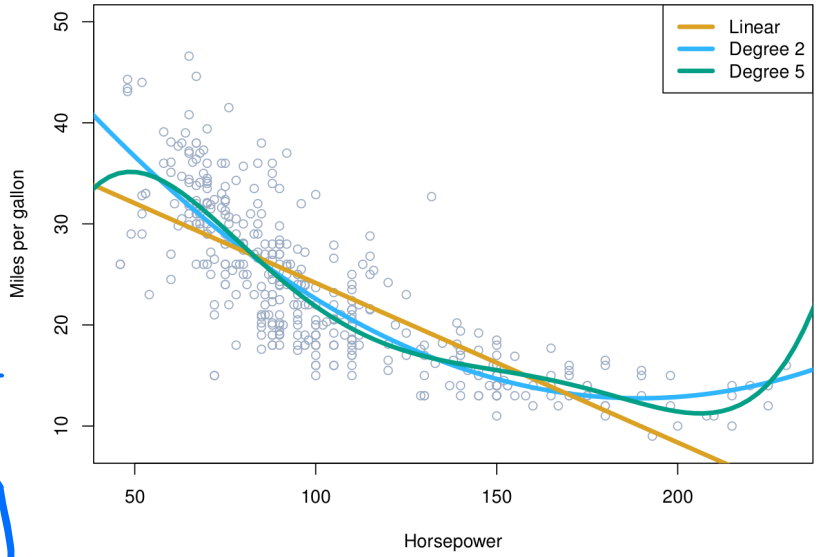
Nonlinear relationships

student compared to nonstudents

$$\texttt{mpg} = \beta_0 + \beta_1 \cdot \overset{x_1}{\texttt{horsepower}} + \beta_2 \cdot \overset{x_2}{\texttt{horsepower}}^2 + \varepsilon$$

- Still tech linear because we do multi-linear regression where $x_1 = hp$ and $x_2 = hp^2$
- adding extra terms (with higher powers) could cause overfitting

This is called polynomial regression



	Coefficient	Std. error	t-statistic	p-value
Intercept	56.9001	1.8004	31.6	< 0.0001
horsepower	-0.4662	0.0311	-15.0	< 0.0001
horsepower^2	0.0012	0.0001	10.1	< 0.0001

Practice and Question Time!

For the remaining part of class, work on the jupyter notebook in groups. If you have any lingering questions or confusion, this is a great time to call me over as well!

Next time

Lec #	Date			Reading
1	Mon	Aug 28	Intro / First day stuff / Python Review Pt 1	1
2	Wed	Aug 30	What is statistical learning?	2.1
	Fri	Sep 1	Assessing Model Accuracy	2.2.1, 2.2.2
3	Mon	Sep 4	No class - Labor day	
4	Wed	Sep 6	Linear Regression	3.1
5	Fri	Sep 8	More Linear Regression	3.1/3.2
6	Mon	Sep 11	Even more linear regression	3.2.2
7	Wed	Sep 13	Probably more linear regression	3.3
8	Fri	Sep 15	Intro to classification, Logisitic Regression	2.2.3, 4.1, 4.2, 4.3
9	Mon	Sep 18	More logistic regression	
10	Wed	Sep 20	Multiple Logistic Regression / Multinomial Logistic Regression	
11	Fri	Sep 22	Overflow/Project day?	
	Mon	Sep 25	***Review***	
	Wed	Sep 27	**Midterm #1**	
	Fri	Sep 29	No class - Dr Munch out of town	
12	Mon	Oct 2	Leave one out CV	5.1.1, 5.1.2
13	Wed	Oct 4	k-fold CV	5.1.3
14	Fri	Oct 6	More k-fold CV,	5.1.4-5
15	Mon	Oct 9	k-fold CV for classification	5.1.5
16	Wed	Oct 11	Resampling methods: Bootstrap	5.2
17	Fri	Oct 13	Subset selection	6.1